

Reed Putnam Practice: Semester Plan, Fall 2026

Coordinator: Kyle Ormsby · Sessions Thu 9:10–10:00 and Fri 1:40–2:30

The exam. Saturday, December 5, 2026 (87th Putnam). New in 2026: four 90-minute sessions with designated breaks, replacing the old two 3-hour sessions. Students bring blue/black pens and the Putnam ID number from the registration platform; no calculators, computers, references, or drawing tools; no leaving and re-entering during a session. Supervisor registration closes November 15; student registration stays open until the competition begins.

The sessions. Nine weeks, two identical 50-minute sessions per week (Thursday 9:10 and Friday 1:40). Each session: ~15-minute mini-lecture on one technique (instructor notes provided), then ~35 minutes of group work on a six-problem handout. Problems 1–2 are warm-ups below Putnam level, 3–4 are A1/B1-level Putnam problems, 5–6 are A3–B5-level take-home problems. Solutions are posted after the Friday session.

Calendar constraints. Fall break Oct 17–25 (no session Oct 22/23). Thanksgiving Nov 26–29 (no session Nov 26/27). Last day of classes Dec 9; the final session (Dec 3/4) is the week of the exam.

Schedule

Wk	Dates	Technique (mini-lecture)	Handout problems (warm-ups; easy Putnam; take-home)
1	Thu Sep 24 Fri Sep 25	Induction, pigeonhole, and the extremal principle; how to write a solution that scores	odd-parts pigeonhole; Erdős–Szekeres; Putnam 2006 B2, 2012 A1; Putnam 2016 B3, 2002 A5
2	Thu Oct 1 Fri Oct 2	Invariants, monovariants, parity, and coloring	domino/tetromino tilings; $ a - b $ and sign-flip processes; Putnam 2024 B1, 2023 B1; Putnam 2008 A3, 2016 A4
3	Thu Oct 8 Fri Oct 9	Polynomials: roots, Vieta, $a - b \mid p(a) - p(b)$, interpolation, growth	integer-polynomial divisibility; Lagrange interpolation; Putnam 2023 A2, 2024 A2; Putnam 2018 B2, 2014 B4
4	Thu Oct 15 Fri Oct 16	Number theory: congruences, orders, Fermat/Euler/Wilson, valuations	$30 \mid n^5 - n$; Mersenne factors and orders; Putnam 2018 A1, 2016 A1; Putnam 2018 B3, 2023 B5
<i>Fall break Oct 17–25: no session Oct 22/23.</i>			
5	Thu Oct 29 Fri Oct 30	Combinatorics: bijections, double counting, recursions, generating functions	no-two-consecutive subsets; Mantel’s theorem; Putnam 2003 A1, 2005 A2; Putnam 2005 B4, 2015 B5
6	Thu Nov 5 Fri Nov 6	Inequalities and analysis: AM–GM, Cauchy–Schwarz, Jensen, integrals, sequences, MVT	AM–GM and Nesbitt; an MVT/IVT classic; Putnam 2003 A2, 2014 B2; Putnam 2012 B4, 2023 A3
7	Thu Nov 12 Fri Nov 13	Linear algebra: determinants, eigenvalues, trace, rank, structured matrices	$A^2 = -I$ and $AB - BA = I$; the $aI + b(J - I)$ determinant; Putnam 2014 A2, 2019 B3; Putnam 2018 A2, 2021 B5

Wk	Dates	Technique (mini-lecture)	Handout problems (warm-ups; easy Putnam; take-home)
8	Thu Nov 19 Fri Nov 20	Probability, expectation, and games: indicators, symmetry, conditioning, pairing and strategy stealing	fixed points and records; coins on a round table; Putnam 2022 A4, 2002 B4; Putnam 2023 B3, 2024 B4
<i>Thanksgiving Nov 26–29: no session Nov 26/27.</i>			
9	Thu Dec 3 Fri Dec 4	Exam week: format, triage, time budget, writing for points; technique checklist	Timed 35-minute mini-set: Putnam 2013 A1, 2024 A1, 2020 A2; take-home: Putnam 2021 A3, 2023 A4, 2019 A3

Running a session

0:00–0:15, mini-lecture. Talk from `notes.pdf`: state the toolbox facts, do the two or three worked examples on the board (they are deliberately not on the handout), and end with the “how to recognize it” paragraph. The notes include a suggested minute-by-minute split.

0:15–0:45, group work. Hand out `handout.pdf`. Groups of two or three; aim for Problems 1–4. Circulate; when a group is stuck on a warm-up for more than five minutes, give the first sentence of the solution. Problems 5–6 are take-home and are meant to be hard.

0:45–0:50, debrief. One group presents the key idea (not the full write-up) of Problem 3 or 4. Remind everyone that solutions post after Friday.

Between weeks. Post `solutions.pdf` after the Friday session. Students who want more can work the corresponding topic handouts from CMU (Po-Shen Loh) and Stanford (Ravi Vakil) linked in the announcement email.

Design notes

The order of topics runs from low-machinery, high-cleverness techniques (Weeks 1–2), so that first-years can play immediately, through the workhorse content areas (Weeks 3–7), to probability and games (Week 8), which draws on everything before it. Week 9 is a rehearsal for the new four-session format rather than new content. Every Putnam problem on the handouts was verified against Kiran Kedlaya’s archive (kskedlaya.org/putnam-archive) for year, label, statement, and answer, and no problem appears in more than one week. Problems are rated ★ (warm-up) to ★★★ (hard Putnam).

Difficulty calibration is intentionally generous at the low end: a strong first-year should finish Problems 1–2 in ten minutes, and the A1/B1-level problems are drawn from the years in which they were among the most-solved problems. If a group is racing ahead, send them to Problem 5.

Files

`weekNN/notes.pdf` (instructor lecture notes), `weekNN/handout.pdf` (student handout), `weekNN/solutions.pdf` (full solutions with key ideas and remarks), with LaTeX sources alongside. All files share `common/preamble.tex`; edit that file to change the look of everything at once.

Loose ends to fill in

Room for the practice sessions and for the December 5 exam (the Week 9 notes and handout carry **ROOM TBD** placeholders); the problem-to-session split for the four 90-minute blocks once the proctor packet arrives (Week 9 notes, [**SPLIT TBD**]); the student registration link from the MAA, to be sent to everyone who replied.